

Practical 4 : Digital Signature

AIM : Implement digital signature algorithms such as RSA-based signatures, and verify the integrity and authenticity of digitally signed messages.

CODE :

```
print(" T120 DHEERAJ SINGH")
print()
from cryptography.hazmat.primitives.asymmetric import rsa, padding
from cryptography.hazmat.primitives import hashes
from cryptography.exceptions import InvalidSignature

private_key = rsa.generate_private_key(
    public_exponent=65537,
    key_size=2048
)

public_key = private_key.public_key()

message = input("Enter message: ").encode()

signature = private_key.sign(
    message,
    padding.PSS(
        mgf=padding.MGF1(hashes.SHA256()),
        salt_length=padding.PSS.MAX_LENGTH
    ),
    hashes.SHA256()
)

print("\nSignature:", signature.hex())

try:
    public_key.verify(
        signature,
        message,
        padding.PSS(
            mgf=padding.MGF1(hashes.SHA256()),
            salt_length=padding.PSS.MAX_LENGTH
        ),
        hashes.SHA256()
```

```
)
print("Signature verified successfully.")
except InvalidSignature:
    print("Signature verification failed.")

modified_message = input("\nEnter message to verify: ").encode()

try:
    public_key.verify(
        signature,
        modified_message,
        padding.PSS(
            mgf=padding.MGF1(hashes.SHA256()),
            salt_length=padding.PSS.MAX_LENGTH
        ),
        hashes.SHA256()
    )
    print("Message is authentic and unchanged.")
except InvalidSignature:
    print("Message is modified or signature is invalid.")
```

OUTPUT :

```
IDLE Shell 3.14.6
File Edit Shell Debug Options Window Help
Python 3.14.6 (tags/v3.14.6:c63aec6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>>
= RESTART: C:/Users/IT-51/Downloads/T120 DHEERAJ SINGH 4th PRACTICAL CYBER SECURITY.py
T120 DHEERAJ SINGH

Enter message: Transfer Rs.10000 to Account xxxx.xxxx.xxxx.5914

Signature: b237d4002c09f10b5e4174b3c8a001cb9c171c128549e9ba38b3f49e4886e99184a897796bb4b5219c05486da72fefb73e85831e3b007435b503a7
b4373f498d49da35419cdeaf14e98398ca4b27692e123d3c95fd00ca6393976904ca24c03e4e792e8b5ee66641d3a8778c5d0af9c1c193d41d9fa3ca1231b4774
220638d9adfc82cd850774716c3efcba4986208f6262f4006529876782984dce618be4ada84bc4fa6d56f5d32c6567238040aa534487482331c97efc60f110a11
92a13eed7371af1d68df91ad6158f5b67ed43b595a78075a03c0835a61a25aff69a634b9bfc10be59ceab4e747305540859f097ff2aebac741ae8e0efb615bb91
50efe4f
Signature verified successfully.

Enter message to verify: Transfer Rs.10000 to Account xxxx.xxxx.xxxx.5914
Message is authentic and unchanged.

>>>
===== RESTART: C:/Users/IT-51/Downloads/T120 DHEERAJ SINGH 4th PRACTICAL CYBER SECURITY.py =====
T120 DHEERAJ SINGH

Enter message: Transfer Rs.10000 to Account xxxx.xxxx.xxxx.5914

Signature: afbd4f67481268a7df96ad3e935d4d3f939f9be3b27622e5a866b9f7ed7035899b8729301220c64e688da86eaea7fa28819f278d438eb07e55c187c
bbe93c585b075056b8f16876d35facc0fa2a099bb788bcc6a2cc47aa8f59654358bb59af9da1736611c2e4aadd50f329b17550cdc0605caa72e72407167779cca
e6b4e2d3cbe5a77967827ce7db621cc07f25c285b89458a0a9c854808ff356a9689791f8212282eafd6f45ba204e10ad6dc639fc99e4c8abf20d76e8b58972dc2
bdb3ec40fbb9ff2a7f4f1d4f0ecc27812952e64e9ba1789270364fbd2271ecd0f25a9a79ea9581d6f517376a6967948a4829a917a17676c63e8e37504e1dc70d
63a1622
Signature verified successfully.

Enter message to verify: Transfer Rs.50000 to Account xxxx.xxxx.xxxx.5914
Message is modified or signature is invalid.

>>>
```